

TRIVORTEX Research Compendium

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TRIVORTEX Research Program v1.0.0 · EN edition

Program	TRIVORTEX — research-papers
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Abstract

TRIVORTEX is a research program around the three-body problem. Twelve companion studies — each an executable script, a committed JSON protocol and a bilingual monograph — attack the problem through twelve physical systems, with lasers at the center of ten of them. The table lists the headline verified number of every study; the dossiers that follow quote the full machine-readable protocol of each.

1. The program at a glance

TRIVORTEX is a research program around the three-body problem. Twelve companion studies — each an executable script, a committed JSON protocol and a bilingual monograph — attack the problem through twelve physical systems, with lasers at the center of ten of them. The table lists the headline verified number of every study; the dossiers that follow quote the full machine-readable protocol of each.

#	Study	Directory	Status
TRX-01	Radiation-Pressure Restricted Three-Body Problem (Laser on Dust)	TRX-01-laser-radiation-pressure	PASS
TRX-02	Resonant Three-Wave Interaction (Manleyâ€ “Rowe, Ĩ â) ½Â²â) ¾ Optics)	TRX-02-three-wave-mixing	PASS
TRX-03	Three-Soliton Molecule in a Mode-Locked Fiber	TRX-03-soliton-	PASS

#	Study	Directory	Status
	Laser	molecule	
TRX-04	Photon Fluid: Three Kerr Solitons (Spatial, 2-D)	TRX-04-photon-fluid	PASS
TRX-05	Optical Vortices of Laser Beams and the Point-Vortex Analogy	TRX-05-optical-vortices	PASS
TRX-06	Helium Atom as the Coulomb Three-Body Problem (Classical Trajectory Monte Carlo)	TRX-06-helium-three-body	PASS
TRX-07	The Efimov Effect: Universal Quantum Three-Body Physics	TRX-07-efimov	PASS
TRX-08	Three Laser-Cooled Ions in a Linear Paul Trap: the Table-Top Lagrange Triangle	TRX-08-ion-trap	PASS
TRX-09	The Kirchhoff “Chaplygin Three-Vortex Problem (Classical Anchor)	TRX-09-vortex-trio	PASS
TRX-10	Kozai “Lidov Oscillations in Hierarchical Triples	TRX-10-kozai-lidov	PASS
TRX-11	Gravitational Waves from the Figure-Eight Choreography	TRX-11-gw-choreography	PASS
TRX-12	Laser Highway: Light-Sail Stationkeeping at L4/L5	TRX-12-laser-light-sail	PASS

2. Study dossiers

Each dossier quotes the acceptance checks exactly as committed in `research/TRX-*/results/trx*_results.json` — check, measured value, target, tolerance and verdict. Nothing here is retyped by hand: this chapter is assembled by `publications/build/build_special_pdfs.py` directly from the committed protocols.

TRX-01 — Radiation-Pressure Restricted Three-Body Problem (Laser on Dust)

The circular restricted three-body problem (CR3BP) in which primary 1 is illuminated by an intense laser beam. Photon radiation pressure reduces the effective pull of that primary on a massless particle by the factor $(1 - \hat{F}^2)$, with $\hat{F}^2 = F_{\text{rad}} / F_{\text{grav}}$ the “laser-dressed” CR3BP. The laser acts as a third physical agent that reshapes the libration-point landscape without adding a mass.

Study	TRX-01-laser-radiation-pressure
Status	PASS
Full-run time	0.10 s
Code	research/TRX-01-laser-radiation-pressure/code/
Monograph	publications/pdf/TRX-01-laser-radiation-pressure_EN.pdf

Acceptance checks

check	measured	target	tolerance	Status
L1_abscissa_beta0	0.836915	0.836915	5.000e-06	PASS
L4_shift_momentone_in_beta	1	1	1.000e-12	PASS
L4_shift_toward_radiating_primary	1	1	1.000e-12	PASS
L4_max_Reigenvalue_beta0	1.857e-16	0	1.000e-08	PASS
L1_shifts_toward_radiating_primary	1	1	1.000e-12	PASS
Jacobi_drift_L4_orbit_beta0.05	8.882e-16	0	1.000e-10	PASS

TRX-02 — Resonant Three-Wave Interaction (Manley-Rowe, $\hat{A}^2$ Optics)

Resonant three-wave mixing in a lossless, phase-matched $\hat{A}^2$ crystal the canonical “three-body problem of nonlinear optics”. A pump wave at $\hat{A}(t) = \hat{A}(t) + \hat{A}(t)$ exchanges photons with a signal an “idler pair; the Manley-Rowe relations keep the photon bookkeeping exact, and the equal-coupling

amplitude equations are canonically equivalent to the Euler top $\hat{C}$ – the same integrable family as the Kirchhoff three-vortex problem that underlies Theorem 3.1 of TRIVORTEX.

Study	TRX-02-three-wave-mixing
Status	PASS
Full-run time	0.6 s
Code	research/TRX-02-three-wave-mixing/code/
Monograph	publications/pdf/TRX-02-three-wave-mixing_EN.pdf

Acceptance checks

check	measured	target	tolerance	Status
ManleyRowe_I13_drift	5.107e-15	0	1.000e-10	PASS
ManleyRowe_I23_drift	1.776e-15	0	1.000e-10	PASS
ManleyRowe_I12diff_drift	3.775e-15	0	1.000e-10	PASS
Pump_period_repeatability	4.470e-08	0	1.000e-06	PASS
Pump_revival_error	2.442e-15	0	1.000e-08	PASS
Pump_full_depletion_occurs	1	1	1.000e-12	PASS
SHG_tanh2_conversion_error	2.220e-16	0	1.000e-08	PASS

TRX-03 — Three-Soliton Molecule in a Mode-Locked Fiber Laser

Three ultrashort pulses circulating in a passively mode-locked fiber laser form a phase-locked “soliton molecule”: in the reduced particle picture each pulse is a body with the conservative, phase-dependent pair interaction $V(r, \theta) = C \hat{C}(\theta) e^{\hat{C}(\theta) 2r/L} \hat{C}(\theta) C \hat{C}(\theta) e^{\hat{C}(\theta) 2r/L} \cos 2\theta$. The phase-locked triplet is a genuine optical three-body choreography: its equilibrium spacing s^* is fixed by the three-body balance $F(s) + F(2s) = 0$, is compressed $\hat{C}(\theta) \approx 30\%$ below the pair value $\hat{C}(\theta) = \ln(4/3)$, and supports a breathing normal mode verified by FFT against the Hessian prediction to 0.21 %.

Study	TRX-03-soliton-molecule
Status	PASS
Full-run time	1.2 s (4.7 s with –figures)
Code	research/TRX-03-soliton-molecule/code/
Monograph	publications/pdf/TRX-03-soliton-molecule_EN.pdf

Acceptance checks

check	measured	target	tolerance	Status
pair_equilibrium_numeric_vs_analytic	5.551e-17	0	1.000e-10	PASS
antiphase_pi2_purely_repulsive	1	1	1.000e-12	PASS
molecule_final_spacings_equal	2.220e-16	0	1.000e-08	PASS
molecule_final_spacing_equals_s_star	2.776e-17	0	1.000e-06	PASS
conservative_energy_drift	1.332e-15	0	1.000e-10	PASS
breathing_mode_frequency	0.468825	0.468766	0.00468766	PASS

TRX-04 — Photon Fluid: Three Kerr Solitons (Spatial, 2-D)

Three laser beams co-propagating in a focusing Kerr medium behave as a photon fluid: each beam is a spatial soliton of the nonlinear Schrödinger equation, and pairs of beams exchange conservative two-body forces whose sign is set by the relative phase ϕ (in-phase beams attract, anti-phase beams repel $\hat{V} = \frac{1}{2}U\hat{\psi}^\dagger\hat{\psi}\hat{\psi}^\dagger\hat{\psi}$ with the exponential profile $V(r) = \frac{1}{2}U\hat{\psi}^\dagger\hat{\psi}\hat{\psi}^\dagger\hat{\psi}$). The in-phase triplet realizes an optical Lagrange central configuration: an equilateral beam triangle of side $a = 3$ rotates rigidly at $\dot{\theta} = e^{i\phi} \sqrt{3/2} = 0.223130160$, the exponential-force counterpart of the Newtonian $\dot{\theta}^2 = 3Gm/a\hat{A}^3$ of Theorem 3.1. The study verifies the choreography to machine precision: the triangle stays equilateral to $1.8e-08$ over three full rotations, the measured rotation rate matches the analytic law to $1.1\hat{A} \times 10^{-8}$, energy and angular momentum are conserved to $4.7e-16$ and $2.2e-15$, the anti-phase trio expands cleanly to 23.476, and an in-phase binary stays bound with a precessing orbit $\hat{A} = 1$ an optical three-body scattering table closed by 9/9 checks PASS.

Study	TRX-04-photon-fluid
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Status	PASS
Full-run time	9.4 s (9.415 s recorded with –figures)
Code	research/TRX-04-photon-fluid/ code/
Monograph	publications/pdf/TRX-04-photon- fluid_EN.pdf

Acceptance checks

check	measured	target	tolerance	Status
triangle_eq uilateral_de viation	4.441e-15	0	1.000e-06	PASS
measured_r otation_rate	0.22313	0.22313	1.000e-08	PASS
energy_con servation_r otation	2.220e-16	0	1.000e-10	PASS
angular_mo mentum_co nservation	8.882e-16	0	1.000e-10	PASS
antiphase_n o_collapse	1	1	1.000e-12	PASS
antiphase_e xpands	1	1	1.000e-12	PASS
energy_con servation_r epulsion	3.053e-16	0	1.000e-10	PASS
binary_stay s_bound	1	1	1.000e-12	PASS
binary_ener gy_conserv ation	1.771e-12	0	1.000e-10	PASS

TRX-05 — Optical Vortices of Laser Beams and the Point-Vortex Analogy

Three phase singularities $\hat{\mathbf{c}}$ optical vortices $\hat{\mathbf{c}}$ of a paraxial laser field behave, to leading order, exactly like Kirchhoff point vortices of ideal fluid dynamics. The study realizes the vortex representation of TRIVORTEX *literally in light*: three same-sign singularities of the field $\hat{\mathbf{c}}(\mathbf{z}) = \exp(\hat{\mathbf{c}}_r \hat{\mathbf{r}}^2/w\hat{\mathbf{a}}^2)\hat{\mathbf{c}} \cdot \hat{\mathbf{r}}(z \hat{\mathbf{a}}_r^2/w\hat{\mathbf{a}}^2 z_k)$ sit on an equilateral triangle of side $a = 1$ and rotate rigidly with the analytic angular velocity $\hat{\mathbf{c}} = 3\hat{\mathbf{c}}_r/(2\hat{\mathbf{c}}_r a\hat{\mathbf{a}}^2) =$

0.477464829, measured to 1e-8, while the angular impulse $I = \int \mathbf{r} \times \mathbf{p} \cdot d\mathbf{r}$ (the many-vortex prototype of the Chaplygin integral) stays pinned at $a\hat{A}^2 = 1$ and the winding number of the rendered field equals 3 exactly â€” the analogue of orbital angular momentum $3\hat{a}_{\text{photon}}$ per photon.

Study	TRX-05-optical-vortices
Status	PASS
Full-run time	4.83 s (4.827 s recorded with –figures)
Code	research/TRX-05-optical-vortices/code/
Monograph	publications/pdf/TRX-05-optical-vortices_EN.pdf

Acceptance checks

check	measured	target	tolerance	Status
rotation_rate_vs_analytic	0.477465	0.477465	1.000e-08	PASS
angular_impulse_I_conserved	1.332e-15	0	1.000e-12	PASS
kirchhoff_hamiltonian_conserved	3.534e-16	0	1.000e-12	PASS
field_minimum_track_vortices	0.016125	0	0.06	PASS
total_winding_number_is_3	3	3	1.000e-12	PASS

TRX-06 — Helium Atom as the Coulomb Three-Body Problem (Classical Trajectory Monte Carlo)

The helium atom â€” a nucleus of charge $Z = 2$ and two electrons â€” is the smallest physical system whose classical limit is the honest three-body problem: two attractive and one repulsive Coulomb pair, all of order unity. The study runs a deterministic classical trajectory Monte Carlo (CTMC) ensemble of 3200 trajectories launched in the Wannier configuration â€” a radially staggered chain at hyperradius $R\hat{A} = 3$ on the exact energy shell, with a fixed kick $\hat{D} = 0.01$ and a documented softening $\hat{\mu} = 0.1$ a.u. â€” and measures the autoionization channel: one electron is captured while the other escapes carrying on average 7.28â€” the excess energy, the fingerprint of three-body energy transfer through the eâ€” »â€” “eâ€” » repulsion. The double-escape statistics are read against Wannier’s threshold law $P_{\text{DE}} \hat{a}_{\text{threshold}} E^{1.056}$, reported honestly for the strong-coupling regime of the fixed-launch geometry.

Study	TRX-06-helium-three-body
Status	PASS
Full-run time	37.7 s
Code	research/TRX-06-helium-three-body/code/
Monograph	publications/pdf/TRX-06-helium-three-body_EN.pdf

Acceptance checks

check	measured	target	tolerance	Status
max_relative_energy_difference_valid	2.884e-06	0	4.000e-04	PASS
outcome_classes_sum_to_N	120	120	1.000e-12	PASS
autoionization_channel_active	1	1	1.000e-12	PASS
escaping_electron_energy_overshoot	1	1	1.000e-12	PASS
fit_reproducible	1	1	1.000e-12	PASS

TRX-07 — The Efimov Effect: Universal Quantum Three-Body Physics

Three identical bosons with resonant (unitary) two-body interactions form an infinite Rydberg-like series of bound three-body states (Efimov trimers) even though the pair potential binds no dimer. The spectrum is universal: it is governed by a single transcendental exponent s_0 defined by $s_0 \hat{A} \cdot \cosh(\sqrt{3} s_0/2) = (8/\hat{A}^3) \hat{A} \cdot \sinh(\sqrt{3} s_0/6)$, $s_0 = 1.0062378$ (target 1.0062458), and it is organized in a geometric ladder: trimer sizes grow by $\exp(\sqrt{3}/s_0) = 22.694383$ and energies drop by $\exp(2\sqrt{3}/s_0) = 515.035001$ per rung. The study realizes the ladder numerically on the hyperradial adiabatic potential $(s_0 \hat{A}^2 \hat{A}^{1/4})/\hat{R}^2$: the four computed trimers reproduce the universal energy ratio within $\hat{A} \pm 0.1\%$ ($515.509 / 514.964 / 514.963$), and a three-body-parameter sweep shows the ratios invariant (spread $1.9 \hat{A} \cdot 10^{-6} \gg \hat{A}^{1/4}$) while absolute energies follow the exact $R \hat{A} \gg \hat{A}^2$ power law (the quantum twin of the scale-invariant vortex triangle of TRIVORTEX).

Study	TRX-07-efimov
Status	PASS

Full-run time	5.77 s (5.771 s recorded with –figures)
Code	research/TRX-07-efimov/code/
Monograph	publications/pdf/TRX-07-efimov_EN.pdf

Acceptance checks

check	measured	target	tolerance	Status
s0_transcendental_root	1.00624	1.00625	1.000e-05	PASS
efimov_length_ratio	22.6944	22.7	0.05	PASS
efimov_energy_ratio	515.035	515.03	0.05	PASS
spectrum_all_negative	1	1	1.000e-12	PASS
ladder_ratio_E0_over_E1	515.477	515.035	180.262	PASS
ladder_ratio_E1_over_E2	514.932	515.035	180.262	PASS

TRX-08 — Three Laser-Cooled Ions in a Linear Paul Trap: the Table-Top Lagrange Triangle

Three laser-cooled ions in the transverse plane of a linear Paul trap form a table-top three-body problem: a two-dimensional harmonic rf pseudopotential confines them, their mutual Coulomb repulsion pushes them apart, and Doppler cooling (modelled as a linear drag $\mathbf{F}_{\text{drag}} = -\gamma \mathbf{v}$) dissipates energy until the ions crystallize into the equilateral Lagrange triangle of side $a = (3\hbar/I)^{1/3} = (3\hbar/m\nu)^{1/3} = 1.4422495703$. The study verifies the whole chain to machine precision: sides equal to $4.4\text{e-}16$, central angles $2\pi/3$ to $4.4\text{e-}16$, the normal-mode ladder $\{0, 1\}$ at $(\tilde{A}^2)^{1/2}$, $\{1, 2\}$ at $(3\tilde{A}^2)^{1/2}$, $\{2, 3\}$ at $(5\tilde{A}^2)^{1/2}$, $\{3, 4\}$ at $(7\tilde{A}^2)^{1/2}$ confirmed to $1.4\cdot 10^{-16}$, and a conservative hold at the exact equilibrium that keeps the crystal static to $3.9\text{e-}15$ over fifty time units with energy drift $8.9\text{e-}16$ the trapped-ion twin of the Lagrange relative equilibrium behind TRIVORTEX Theorem 3.1 and the smallest ion-crystal quantum simulator.

Study	TRX-08-ion-trap
Status	PASS
Full-run time	6.79 s (6.794 s recorded with –figures)
Code	research/TRX-08-ion-trap/code/
Monograph	publications/pdf/TRX-08-ion-trap_EN.pdf

Acceptance checks

check	measured	target	tolerance	Status
global_minimum_reached	0	0	1.000e-09	PASS
crystallization_energy_released	1	1	1.000e-12	PASS
crystal_side_equality_analytic	1.44225	1.44225	1.000e-08	PASS
crystal_equilateral	0	0	1.000e-08	PASS
crystal_angles_120deg	4.441e-16	0	1.000e-06	PASS
zero_rotation_mode	1	1	1.000e-12	PASS
two_com_modes_at_omega0	1	1	1.000e-12	PASS
breathing_mode_sqrt3	1	1	1.000e-12	PASS
static_equilibrium_holds	1.388e-15	0	1.000e-09	PASS
conservative_energy_conserved	8.882e-16	0	1.000e-10	PASS

TRX-09 — The Kirchhoff (“Chaplygin Three-Vortex Problem (Classical Anchor)

The classical point-vortex anchor of TRIVORTEX: three Kirchhoff vortices with circulations $\hat{\Gamma}_1, \hat{\Gamma}_2, \hat{\Gamma}_3$ and the angular impulse $I = \hat{\Gamma}_1 \hat{\Gamma}_2 r | \hat{A}^2 \hat{a}_{\text{PACH}}$ the prototype of the Chaplygin topological integral C_{Ch} of Theorem 3.1. Two canonical regimes are verified numerically: the same-sign equilateral triangle rotates rigidly with $\hat{\Gamma} = 3\hat{\Gamma}_1/(2\hat{\Gamma}_1 \hat{a}_{\text{PAC}} \hat{A}^2) = 0.477464829$ (the vortex Lagrange solution), and the mixed-sign $(1, 1, \hat{a}_{\text{HTSPUE}}^{-1})$ trio launched from the right-isosceles configuration sits exactly on the Aref collapse manifold $I = H = P = Q = 0$ and evolves non-rigidly while all four invariants stay pinned to machine precision.

Study	TRX-09-vortex-trio
Status	PASS
Full-run time	0.54 s (7.389 s recorded with –figures)
Code	research/TRX-09-vortex-trio/ code/

Monograph	publications/pdf/TRX-09-vortex-trio_EN.pdf
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Acceptance checks

check	measured	target	tolerance	Status
rotation_rate_3 G_over_2pi_a2	0.477465	0.477465	1.000e-08	PASS
triangle_stays_ equilateral	3.331e-16	0	1.000e-08	PASS
angular_impuls e_I_conserved	8.882e-16	0	1.000e-12	PASS
hamiltonian_co nserved	2.474e-16	0	1.000e-12	PASS
linear_impulse _conserved	7.216e-16	0	1.000e-12	PASS
aref_collapse_c onditions_hold	1.388e-17	0	1.000e-12	PASS
mixed_sign_inv ariants_conser ved	1.310e-14	0	1.000e-11	PASS
mixed_sign_sh ape_evolves	1	1	1.000e-12	PASS

TRX-10 — Kozai “Lidov Oscillations in Hierarchical Triples

The Kozai “Lidov mechanism: a test particle on an inclined inner orbit of a hierarchical triple, perturbed by a distant companion, periodically exchanges eccentricity for inclination while the z-angular momentum $j_z = \sqrt{1 - e^2} \cos i$ stays constant; for a nearly circular initial orbit the eccentricity climbs to the closed-form maximum $e_{\max} = \sqrt{1 - (5/3) \cos^2 i_0}$ whenever i_0 exceeds the Kozai angle 39.23° . The study refuses to code the memorized result: the doubly-averaged quadrupole potential is built numerically $\hat{C}_2$ orbit quadrature, (e, i) tabulation, cubic-spline Hamiltonian $\hat{C}_2$ and the resulting flow is audited against analytic benchmarks and an independent direct integration of the full three-dimensional restricted problem.

Study	TRX-10-kozai-lidov
Status	PASS
Full-run time	64.6 s
Code	research/TRX-10-kozai-lidov/ code/
Monograph	publications/pdf/TRX-10-kozai-

Acceptance checks

check	measured	target	tolerance	Status
e_max_i0_60	0.763752	0.763763	0.002	PASS
e_max_i0_70	0.897229	0.897239	0.002	PASS
kl_period_halves_with_m3	0.493173	0.5	0.03	PASS

TRX-11 — Gravitational Waves from the Figure-Eight Choreography

The figure-eight choreography (Chenciner & Montgomery 2000) is three equal masses chasing each other along a single closed curve with zero angular momentum is treated as a laboratory gravitational-wave source. In the quadrupole approximation ($G = c = D = 1$) the mass quadrupole $Q_{ij} = \frac{1}{2} m_k x_{ki} x_{kj}$ drives a plus-polarised strain that repeats six times per orbit period $T = 6.325914$, and the choreography symmetry locks the spectrum into a pure harmonic comb with the dominant line at $n = 6$ of the orbital comb.

Study	TRX-11-gw-choreography
Status	PASS
Full-run time	8.17 s
Code	research/TRX-11-gw-choreography/code/
Monograph	publications/pdf/TRX-11-gw-choreography_EN.pdf

Acceptance checks

check	measured	target	tolerance	Status
period_closure_error	1.232e-08	0	5.000e-08	PASS
period_in_expected_range	1	1	1.000e-12	PASS
energy_conserved	3.331e-15	0	1.000e-12	PASS
angular_momentum_is_	1.443e-15	0	1.000e-09	PASS

check	measured	target	tolerance	Status
zero				
luminosity_ finite_positi ve	1	1	1.000e-12	PASS
dominant_h armonic_on _comb	0.00149963	0	0.05	PASS

TRX-12 — Laser Highway: Light-Sail Stationkeeping at L4/L5

A photon light sail in the Earth-Moon circular restricted three-body problem, pushed by a 1 MW laser beam: the thrust $a_L = 2P/(cm)$ is pointed by a proportional-derivative beam-steering law and capped by the photon ceiling $a_{max} = 0.2443$ (dimensionless). The study demonstrates the two core operations of a “laser highway” — holding a spacecraft at the linearly stable but neutrally drifting L4 point to $7.5e-07$ of the nominal position, and pumping the Jacobi constant down with along-velocity thrust to move between invariant manifolds.

Study	TRX-12-laser-light-sail
Status	PASS
Full-run time	41.4 s
Code	research/TRX-12-laser-light-sail/ code/
Monograph	publications/pdf/TRX-12-laser- light-sail_EN.pdf

Acceptance checks

check	measured	target	tolerance	Status
controlled_l 4_bound	8.763e-05	0	2.000e-04	PASS
control_sat uration_frac tion	0	0	0.05	PASS
free_drift_n ever_conver ges	1	1	1.000e-12	PASS
jacobi_pum ped_monoto nically	1	1	1.000e-12	PASS

check	measured	target	tolerance	Status
power_table _consistent	1	1	1.000e-12	PASS

4. Reproduction

smoke: all twelve acceptance suites in seconds (what CI runs)

for d in research/TRX-*/code/*.py; **do** python3 "\$d" --smoke; **done**

full: every study, full statistics, JSON protocols

for d in research/TRX-*/code/*.py; **do** python3 "\$d"; **done**

ultra-resolution figure sets (300 dpi)

for d in research/TRX-*/code/*.py; **do** python3 "\$d" --figures; **done**

3. The verification discipline — Every study is a deterministic script (numpy/scipy only) that executes its physics, compares the measured numbers against published or registered targets, and writes a JSON protocol with a full parameter snapshot. The `--smoke` mode runs the same acceptance logic in seconds and is wired into CI; the full mode and the `--figures` mode regenerate the 300-dpi figure set. A study passes only when every check is green — the same discipline the core verification ladder V1–V4 applies to Theorem 3.1 of the vortex document.